

ISC Class XII Maths — Chapter 8

Application of Integrals — Formula Sheet

A. Area Under a Curve — Core Formula

Vertical strips (curve given as $y = f(x)$), area between the curve and the x -axis from $x = a$ to $x = b$:

$$\text{Area} = \int_a^b y \, dx = \int_a^b f(x) \, dx$$

Horizontal strips (curve given as $x = g(y)$), area between the curve and the y -axis from $y = c$ to $y = d$:

$$\text{Area} = \int_c^d x \, dy$$

Choose whichever strip makes the boundary equation simplest to integrate — vertical strips for "bounded by $x = a, x = b$," horizontal strips for "bounded by $y = c, y = d$."

B. Area Between Two Curves

For two curves with $y_{\text{upper}} \geq y_{\text{lower}}$ on $[a, b]$:

$$\text{Area} = \int_a^b (y_{\text{upper}} - y_{\text{lower}}) \, dx$$

If the curves cross inside $[a, b]$, split the integral at each crossing point and always subtract the lower curve from the upper curve on *that* sub-interval.

C. The Signature Standard Integral

$$\int \sqrt{a^2 - x^2} \, dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) + C$$

This single result drives almost every circle and ellipse area problem in this chapter — memorise it exactly, including the $\frac{1}{2}$ and $\frac{a^2}{2}$ coefficients.

D. Standard Curve Equations Reference

Circle (centre origin, radius a): $x^2 + y^2 = a^2 \Rightarrow y = \sqrt{a^2 - x^2}$

Parabola (opens right): $y^2 = 4ax \Rightarrow y = \sqrt{4ax}$
(opens up): $x^2 = 4ay$

Ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow y = \frac{b}{a} \sqrt{a^2 - x^2}$

Line through $(x_1, y_1), (x_2, y_2)$: $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

Worked Pattern: Area of a Circle

For $x^2 + y^2 = a^2$, use symmetry across all four quadrants rather than integrating the full circle directly:

1. Solve for $y = \sqrt{a^2 - x^2}$ (upper half, first quadrant arc).
2. Area of one quadrant = $\int_0^a \sqrt{a^2 - x^2} \, dx$; apply the standard result (Sec. C).
3. Multiply by 4 for the full circle: $\text{Area} = 4 \times \frac{\pi a^2}{4} = \pi a^2$.

Worked Pattern: Area of an Ellipse

For $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$:

1. Solve for $y = \frac{b}{a} \sqrt{a^2 - x^2}$ (first-quadrant arc).
2. Area of one quadrant = $\int_0^a \frac{b}{a} \sqrt{a^2 - x^2} \, dx = \frac{b}{a} \cdot \frac{\pi a^2}{4}$ (standard result, Sec. C).
3. Multiply by 4: $\text{Area} = 4 \times \frac{\pi ab}{4} = \pi ab$ — quote this directly once derived once.

E. Circular Segments

A **segment** is the region cut off by a chord (a line not through the centre).

Minor segment (smaller piece): Area = (Area under the arc, from the chord's x to a) computed directly via Sec. C over the appropriate limits — draw the figure to identify the correct limits of integration.

Alternative: Segment area = Sector area – Triangle area, where Sector = $\frac{1}{2}a^2\theta$ and Triangle = $\frac{1}{2}a^2 \sin \theta$ (θ in radians, the angle subtended at the centre).

Worked Pattern: Line-and-Curve Regions

1. Sketch both the line and the curve — this chapter's marks depend heavily on a correct rough figure.
2. Find point(s) of intersection by solving the two equations simultaneously.
3. Identify which curve is "on top" (y_{upper}) in each relevant sub-interval.
4. Set up Area = $\int (y_{\text{upper}} - y_{\text{lower}}) dx$ over each sub-interval (split at intersection points if the upper/lower curve swaps), and add the pieces.

F. Symmetry Shortcuts

If the region is symmetric about the x -axis: find the area of the **upper half only** and multiply by 2.

If symmetric about **both** axes (full circle/ellipse centred at origin): find the **first-quadrant** area and multiply by 4.

Always state *why* the symmetry holds (e.g. "the curve contains only even powers of y , so it is symmetric about the x -axis") before applying the multiplier — this is a method mark in ISC board answers.

G. Common Traps to Avoid

- Always **sketch the region first** — setting up limits without a figure is the single biggest source of wrong answers in this chapter.
- In Sec. C's standard result, don't drop the $\frac{1}{2}$ and $\frac{a^2}{2}$ coefficients, and don't forget the $\sin^{-1}$ term entirely (a very common slip when rushing).
- When two curves swap which one is "on top," you **must** split the integral at the crossing point — a single integral across the whole interval silently gives the wrong (signed, cancelling) answer.
- Area is always reported as a **positive** quantity — if a computed integral comes out negative, take the absolute value; it means y_{lower} and y_{upper} were swapped.
- Double-check which variable you integrated with respect to matches the strip orientation you sketched (vertical strips $\rightarrow dx$, horizontal strips $\rightarrow dy$).

H. Quick Recognition Guide

Pattern seen	Use
“Area bounded by $y = f(x)$ and lines $x = a, x = b$ ”	Sec. A vertical strips
“Area bounded by ... and $y = c, y = d$ ”	Sec. A horizontal strips
Two curves/lines given, “find the area between/enclosed by”	Sec. B, find intersection first
Circle equation $x^2 + y^2 = a^2$ appears anywhere in the integral	Sec. C standard result
“Find the area of the circle/ellipse” (whole shape)	Worked Pattern: Circle or Ellipse by Symmetry
“Find the area of the (minor/major) segment”	Sec. E
Line <i>and</i> curve both given in the same question	Worked Pattern: Line and Curve
Region symmetric about an axis (even powers only)	Sec. F — halve the work, double/quadruple the result
Answer comes out negative	Sec. G — take absolute value, check which curve is on top